

Slab-Geometry Note: Embedded TI/Trivial Interface and Local Chern Marker

April 22, 2026

1 Purpose

This note records the setup and current numerical results for the slab/interface calculation implemented in `src/interface_chern`. The geometry is a finite explicit slab

$$-M \leq y \leq M - 1 \quad (1)$$

embedded between two semi-infinite bulk regions:

$$y \geq M : \text{TI}, \quad y \leq -M - 1 : \text{triv.} \quad (2)$$

Translation invariance is kept in the x direction, so the calculation is done in mixed (k_x, y) representation.

The main observable is the row-resolved local Chern-marker profile. The two results worth keeping as reference are:

- the equilibrium $M = 100$ slab develops a sharp interface crossover from a trivial-side marker near 0 to a TI-side plateau near -0.81 ,
- in the current monitored setup, a single monitored row at $y = 0$ already collapses the marker profile to nearly zero for all scanned $\Gamma > 0$.

2 Geometry and Hamiltonian

The bulk lattice Hamiltonian is the same Chern-insulator model used in `monitored_CI2.tex`:

$$h_r(k_x, k_y) = t_0 \left[\sigma^1 \sin k_x + \sigma^2 \sin k_y + \sigma^3 (r - \cos k_x - \cos k_y) \right]. \quad (3)$$

We take

$$r_{\text{TI}} = 1, \quad r_{\text{triv}} = 3, \quad (4)$$

so the upper bulk is topological and the lower bulk is trivial.

In mixed (k_x, y) representation the explicit slab Hamiltonian is

$$H_{\text{strip}}(k_x) = \begin{pmatrix} h_{-M} & V & 0 & \cdots & 0 \\ V^\dagger & h_{-M+1} & V & \ddots & \vdots \\ 0 & V^\dagger & h_{-M+2} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & V \\ 0 & \cdots & 0 & V^\dagger & h_{M-1} \end{pmatrix}, \quad (5)$$

with

$$h_y(k_x) = t_0 \left[\sigma^1 \sin k_x + \sigma^3 (r_y - \cos k_x) \right], \quad (6)$$

$$V = \frac{\sigma^2}{2i} - \frac{\sigma^3}{2}, \quad (7)$$

and sharp interface profile

$$r_y = \begin{cases} r_{\text{triv}}, & -M \leq y \leq -1, \\ r_{\text{TI}}, & 0 \leq y \leq M-1. \end{cases} \quad (8)$$

3 Lead Embedding

The explicit slab is not given hard walls. Instead, the boundary rows are coupled to semi-infinite equilibrium leads through embedding self-energies:

$$G_{\text{strip}}^R(\omega, k_x) = \left[(\omega + i\eta) \mathbf{1} - H_{\text{strip}}(k_x) - \Sigma_{\text{TI}}^R(\omega, k_x) - \Sigma_{\text{triv}}^R(\omega, k_x) - \Sigma_M^R \right]^{-1}. \quad (9)$$

The lead self-energies are obtained numerically from surface Green's functions using Lopez-Sancho decimation. In equilibrium, with no monitoring,

$$G_{\text{strip}}^K(\omega, k_x) = \text{sgn}(\omega) \left(G_{\text{strip}}^R - G_{\text{strip}}^A \right), \quad (10)$$

so the fixed bulks satisfy zero-temperature FDT exactly.

4 Monitoring Setup

The monitored runs use the following specific choice:

- only the single row $y = 0$ is monitored,
- both σ^3 -basis orbitals on that row are monitored,
- the monitored retarded self-energy is $\Sigma_M^R = -i\Gamma/2$ on the $y = 0$ row block,
- for the large- M sweep we assume the half-filled monitored saddle, $\Sigma_M^K = 0$, and verify the corresponding monitored densities posteriori.

With this assumption the monitored calculation is a single pass rather than a self-consistent iteration. For the scanned $M = 100$ runs, the posterior deviation of the monitored orbital densities from $1/2$ is at most 4.7×10^{-11} , so the half-filling assumption is numerically self-consistent for this setup.

5 Local Chern Marker

The equal-time correlation matrix is

$$C(k_x) = \frac{1}{2} \mathbf{1} - \frac{i}{2} \int \frac{d\omega}{2\pi} G_{\text{strip}}^K(\omega, k_x). \quad (11)$$

The row-resolved marker is evaluated as

$$c(y) = 2\pi \int_{-\pi}^{\pi} \frac{dk_x}{2\pi} \text{tr}_{\text{orb}} \left[\langle y | C(k_x) [\partial_{k_x} C(k_x), [Y, C(k_x)]] | y \rangle \right], \quad (12)$$

with

$$Y_{yy'} = y \delta_{yy'}. \quad (13)$$

For equilibrium runs this is the standard projector-based marker with $C(k_x) = P(k_x)$. For monitored runs it is used as a nonequilibrium local Hall/Chern-marker diagnostic.

6 Alternative Boundary-First Reconstruction ($M = 0$)

An analytically cleaner route is to step back from the explicit slab and first solve only the monitored *boundary* problem at $y = 0$, then reconstruct the Green's function deeper in the bulk from that boundary data. This is most natural for the semi-infinite edge geometry $y \geq 0$, and is the right starting point if the eventual goal is to build the equal-time correlation matrix in a finite window near the edge.

Let $g_{nm}(\omega, k_x)$, with $n, m \geq 0$, denote the Green's function of the *unmonitored* half-space problem, and let the monitoring self-energy be strictly local to the boundary row,

$$\Sigma_M(\omega, k_x) = |0\rangle \hat{\Sigma}_M(\omega, k_x) \langle 0|, \quad (14)$$

where $\hat{\Sigma}_M$ is a 2×2 matrix in orbital space. The full Green's function satisfies the contour Dyson equation

$$G = g + g \Sigma_M G. \quad (15)$$

Because Σ_M has support only on $y = 0$, this equation closes exactly into a boundary T -matrix problem:

$$G_{nm} = g_{nm} + g_{n0} T g_{0m}, \quad (16)$$

with

$$T = (1 - \hat{\Sigma}_M g_{00})^{-1} \hat{\Sigma}_M = \hat{\Sigma}_M (1 - g_{00} \hat{\Sigma}_M)^{-1}. \quad (17)$$

For the retarded and advanced components,

$$G_{nm}^{R/A} = g_{nm}^{R/A} + g_{n0}^{R/A} T^{R/A} g_{0m}^{R/A}, \quad (18)$$

where

$$T^{R/A} = (1 - \hat{\Sigma}_M^{R/A} g_{00}^{R/A})^{-1} \hat{\Sigma}_M^{R/A}. \quad (19)$$

The Keldysh component follows from the Langreth rules,

$$G_{nm}^K = g_{nm}^K + g_{n0}^R T^R g_{0m}^K + g_{n0}^K T^A g_{0m}^A + g_{n0}^R T^K g_{0m}^A, \quad (20)$$

with

$$T^K = (1 + T^R g_{00}^R) \hat{\Sigma}_M^K (1 + g_{00}^A T^A) + T^R g_{00}^K T^A. \quad (21)$$

Equivalently, all mixed and bulk blocks are reconstructed from the unmonitored-half-space propagators $g_{nm}^{R/A/K}$ and the local boundary matrix $T^{R/A/K}$.

Once G_{nm}^K is known, the equal-time correlation matrix in an edge window $0 \leq n, m \leq N_{\text{edge}}$ is

$$C_{nm}(k_x) = \frac{1}{2} \delta_{nm} \mathbf{1}_{\text{orb}} - \frac{i}{2} \int \frac{d\omega}{2\pi} G_{nm}^K(\omega, k_x). \quad (22)$$

This produces the full correlation matrix supported near the edge without ever introducing a large explicit slab.

What information is needed beyond the current lead boundary condition? This point is important. The present embedding code uses only the *surface-surface* Green's function g_{00} of the semi-infinite bulk in order to build the self-energy. That is sufficient to determine the local boundary Green's function G_{00} , but it is *not* sufficient to reconstruct G_{nm} in the bulk. The formulas above show that one needs, in addition,

$$g_{n0}^{R/A/K}, \quad g_{0m}^{R/A/K}, \quad g_{nm}^{R/A/K} \quad (n, m > 0), \quad (23)$$

or, equivalently, the bulk propagation data that generates these objects. Concretely, this means that the current lead boundary condition encodes only the surface self-energy, while the boundary-first reconstruction requires *surface-to-bulk columns* and, for a whole edge window, the restricted half-space block of the unmonitored bulk Green's function.

For a homogeneous nearest-neighbor bulk this extra information is still calculable: it can be generated from the same half-space problem by a transfer matrix or surface-to-bulk recursion. The key warning is simply that the surface self-energy alone does *not* determine the bulk distribution of the correlator. One must supplement the current boundary condition with explicit propagation data into the bulk.

Implementation-ready edge-window formulation

For coding, the cleanest route is not to propagate directly from G_{00} . Instead, choose an edge window

$$E = \{0, 1, \dots, N_{\text{edge}}\} \quad (24)$$

inside the monitored half-space $y \geq 0$, and compute the full Green's function on that window exactly by embedding the *tail* $y \geq N_{\text{edge}} + 1$ back into the last row of the window.

For the single-edge TI problem the half-space bulk is uniform, with

$$h_{\text{bulk}}(k_x) = t_0 \left[\sigma^1 \sin k_x + \sigma^3 (r_{\text{TI}} - \cos k_x) \right], \quad (25)$$

$$V = \frac{\sigma^2}{2i} - \frac{\sigma^3}{2}. \quad (26)$$

The finite edge-window Hamiltonian is the block tridiagonal matrix

$$H_E(k_x) = \begin{pmatrix} h_{\text{bulk}} & V & 0 & \cdots & 0 \\ V^\dagger & h_{\text{bulk}} & V & \ddots & \vdots \\ 0 & V^\dagger & h_{\text{bulk}} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & V \\ 0 & \cdots & 0 & V^\dagger & h_{\text{bulk}} \end{pmatrix}, \quad (27)$$

whose rows are indexed by $n = 0, \dots, N_{\text{edge}}$.

Let $g_{\text{tail},s}^{R/A}(\omega, k_x)$ denote the surface Green's function of the homogeneous semi-infinite tail $n \geq N_{\text{edge}} + 1$. It satisfies the usual surface Dyson equation

$$g_{\text{tail},s}^R = \left[(\omega + i0^+) \mathbf{1}_{\text{orb}} - h_{\text{bulk}} - V g_{\text{tail},s}^R V^\dagger \right]^{-1}, \quad (28)$$

with $g_{\text{tail},s}^A = (g_{\text{tail},s}^R)^\dagger$. The tail self-energy acting on the last row of the edge window is

$$\Sigma_{\text{tail}}^{R/A/K} = |N_{\text{edge}}\rangle \hat{\Sigma}_{\text{tail}}^{R/A/K} \langle N_{\text{edge}}|, \quad (29)$$

$$\hat{\Sigma}_{\text{tail}}^{R/A} = V g_{\text{tail},s}^{R/A} V^\dagger, \quad (30)$$

$$\hat{\Sigma}_{\text{tail}}^K = \text{sgn}(\omega) \left(\hat{\Sigma}_{\text{tail}}^R - \hat{\Sigma}_{\text{tail}}^A \right), \quad (31)$$

since the bulk tail is fixed in equilibrium at zero temperature.

The exact *unmonitored* Green's function on the entire edge window is then

$$g_E^R = \left[(\omega + i0^+) \mathbf{1} - H_E(k_x) - \Sigma_{\text{tail}}^R \right]^{-1}, \quad (32)$$

$$g_E^A = (g_E^R)^\dagger, \quad (33)$$

$$g_E^K = g_E^R \Sigma_{\text{tail}}^K g_E^A = \text{sgn}(\omega) (g_E^R - g_E^A). \quad (34)$$

This gives all matrix elements $g_{nm}^{R/A/K}$ for $0 \leq n, m \leq N_{\text{edge}}$ in one inversion, with no separate surface-to-bulk recursion.

Monitoring is then added only on the boundary row $n = 0$:

$$\Sigma_M^{R/A/K} = |0\rangle \hat{\Sigma}_M^{R/A/K} \langle 0|. \quad (35)$$

For the monitored problem on the window one may either use the boundary T -matrix formulas above, or simply solve the embedded window Dyson equation directly,

$$G_E^R = \left[(\omega + i0^+) \mathbf{1} - H_E(k_x) - \Sigma_{\text{tail}}^R - \Sigma_M^R \right]^{-1}, \quad (36)$$

$$G_E^A = (G_E^R)^\dagger, \quad (37)$$

$$G_E^K = G_E^R (\Sigma_{\text{tail}}^K + \Sigma_M^K) G_E^A. \quad (38)$$

The correlation matrix supported near the edge is then

$$C_{nm}(k_x) = \frac{1}{2} \delta_{nm} \mathbf{1}_{\text{orb}} - \frac{i}{2} \int \frac{d\omega}{2\pi} [G_E^K(\omega, k_x)]_{nm}, \quad 0 \leq n, m \leq N_{\text{edge}}. \quad (39)$$

For the monitor used in the present code,

$$\hat{\Sigma}_M^R = -\frac{i\Gamma}{2} \mathbf{1}_{\text{orb}}, \quad \hat{\Sigma}_M^A = +\frac{i\Gamma}{2} \mathbf{1}_{\text{orb}}, \quad (40)$$

and, if one keeps the general self-consistent form,

$$\hat{\Sigma}_M^K = -i\Gamma \begin{pmatrix} 1 - 2n_1 & 0 \\ 0 & 1 - 2n_2 \end{pmatrix}, \quad (41)$$

with monitored orbital occupancies extracted from the boundary block

$$n_a = \int_{-\pi}^{\pi} \frac{dk_x}{2\pi} \left[\frac{1}{2} - \frac{i}{2} \int \frac{d\omega}{2\pi} [G_{00}^K(\omega, k_x)]_{aa} \right]. \quad (42)$$

At the half-filled monitored saddle used in the large- M scans,

$$\hat{\Sigma}_M^K = 0, \quad (43)$$

so the full window Keldysh component reduces to

$$G_E^K = G_E^R \Sigma_{\text{tail}}^K G_E^A. \quad (44)$$

Minimal data another agent actually needs. The implementation can therefore be based on the following minimal set of bulk inputs:

1. the local bulk row Hamiltonian $h_{\text{bulk}}(k_x)$,
2. the nearest-row hopping V ,
3. the bulk-tail surface Green's function $g_{\text{tail},s}^{R/A}$,
4. equilibrium FDT for the tail, which fixes Σ_{tail}^K .

No explicit bath distribution matrix beyond equilibrium FDT is needed. What is *not* sufficient is the already-monitored local boundary Green's function G_{00} by itself: one must still rebuild the whole edge window with the tail embedding, because G_{00} alone does not determine the off-diagonal bulk blocks G_{nm} .

7 Numerical Settings Used for the Reference Runs

The large- M runs shown below use

$$M = 100, \quad Nk_x = 41, \quad (45)$$

$$\omega_{\text{dense}} = 2, \quad \omega_{\text{max}} = 6, \quad \Delta\omega_{\text{dense}} = 0.05, \quad \Delta\omega_{\text{coarse}} = 0.15. \quad (46)$$

The code now enforces

$$\eta \geq 2\Delta\omega_{\text{dense}}, \quad (47)$$

so these runs use

$$\eta = 0.1. \quad (48)$$

For the equilibrium $M = 100$ run the diagnostics are

$$\text{fdt_max_error} = 0, \quad \text{lead_residual_max} = 4.56 \times 10^{-14}. \quad (49)$$

8 Results

8.1 Equilibrium slab

The equilibrium $M = 100$ marker profile is shown in Fig. 1. Numerically,

$$\bar{c}_{\text{TI}} = -0.8066, \quad \bar{c}_{\text{triv}} = -0.00556, \quad (50)$$

and representative rows are

$$c(-1) = -0.1036, \quad c(0) = -0.5511, \quad c(1) = -0.8078. \quad (51)$$

So the crossover is localized to the interface region, while the deep TI side quickly saturates to a stable plateau.

The corresponding equilibrium spectral output is shown in Fig. 2. The top panel is the interface-window spectral sum used by the code, while the remaining panels are selected row-resolved spectra from the trivial side, interface region, and TI side. This is the direct spectral companion to the marker profile in Fig. 1.

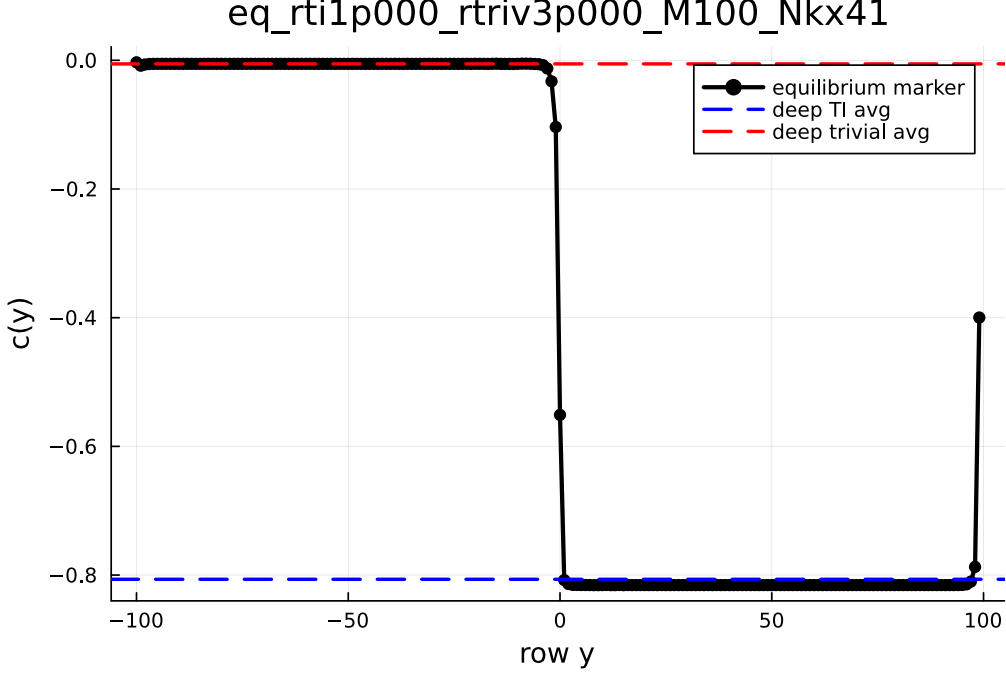


Figure 1: Equilibrium row-resolved Chern marker for the embedded $M = 100$ TI/trivial slab. The trivial side stays near 0, while the TI side develops a plateau near -0.81 .

8.2 Single-row monitoring at $y = 0$

For a single monitored row $y = 0$, monitoring both σ^3 -basis orbitals, the marker profile is strongly suppressed. At $\Gamma = 0.5$, Fig. 3 shows that the profile is already nearly flat, with

$$\bar{c}_{\text{TI}} = -1.21 \times 10^{-4}, \quad \bar{c}_{\text{triv}} = -2.16 \times 10^{-5}, \quad (52)$$

and

$$c(-1) \approx 0, \quad c(0) \approx 0, \quad c(1) \approx 0. \quad (53)$$

The monitored spectral output at the same parameters is shown in Fig. 4. It is useful mainly as a reference diagnostic to compare directly against the equilibrium spectral panels and against the marker collapse in Fig. 3.

8.3 Measurement-rate sweep

The Γ -sweep is shown in Fig. 5. The equilibrium profile is included as the top-left reference panel, while the monitored panels cover

$$\Gamma = 0.05, 0.1, 0.2, 0.5, 1, 2, 5. \quad (54)$$

Within this current setup, the collapse is essentially immediate: every monitored panel is already close to zero for the full scanned range $\Gamma \ll t_0$ through $\Gamma \gg t_0$.

This observation should be interpreted narrowly: it applies to the current choice of monitoring both σ^3 orbitals on a single row with the half-filled monitored saddle enforced. It is not yet evidence that any local monitoring protocol would produce the same behavior.

eq_rti1p000_rtriv3p000_M100_Nkx41 interface sum

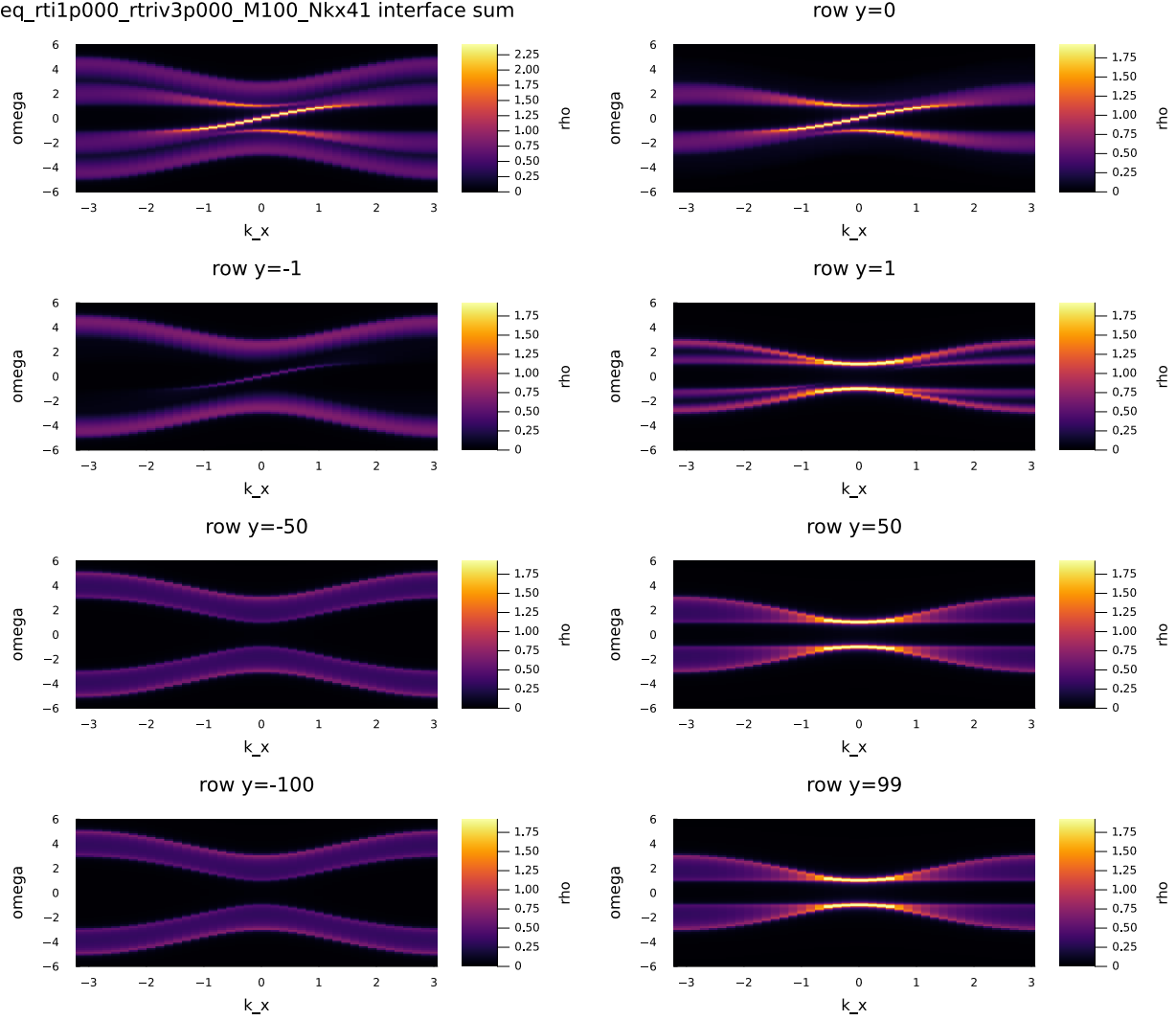


Figure 2: Equilibrium spectral output for the $M = 100$ slab. The first panel is the interface-window sum; the remaining panels are row-resolved spectral functions for selected rows, with $y < 0$ in the left column and $y \geq 0$ in the right column. All row panels share one color scale.

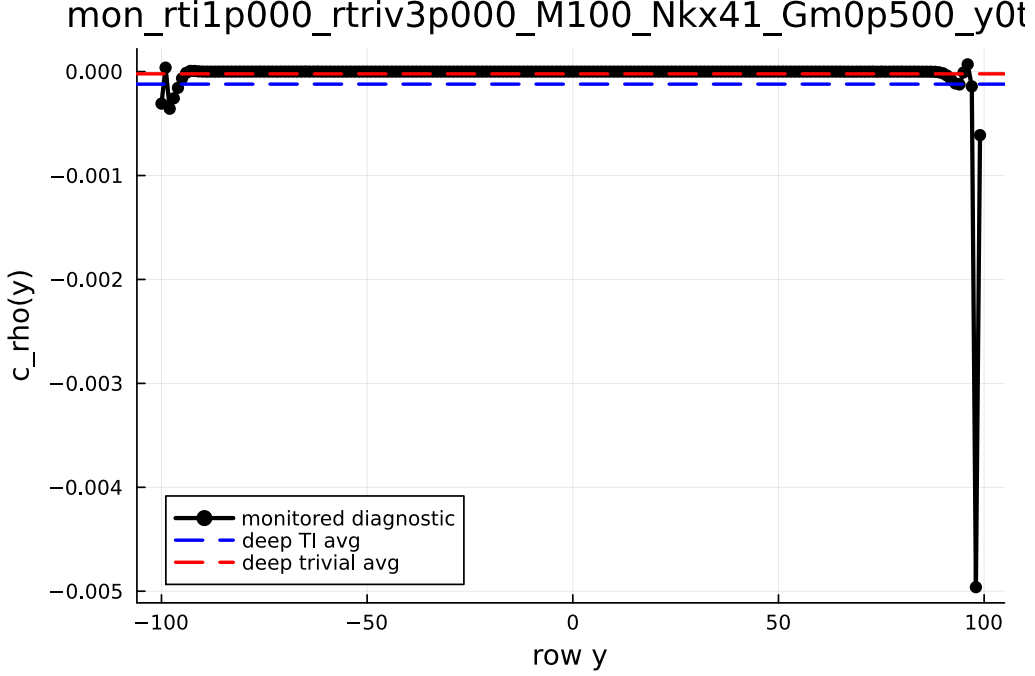


Figure 3: Monitored $M = 100$ marker profile for a single monitored row $y = 0$ with $\Gamma = 0.5$, monitoring both σ^3 -basis orbitals and imposing the half-filled saddle $\Sigma_M^K = 0$. The diagnostic collapses to nearly zero across the entire slab.

9 Takeaway

For the embedded $M = 100$ slab, the equilibrium calculation shows a clear interface-localized Chern-marker crossover. In contrast, the present monitored implementation — single monitored row, both σ^3 orbitals, half-filled monitored saddle — removes that profile almost completely for all scanned nonzero Γ .

The next natural follow-up calculations are:

- monitor only one orbital projector instead of both orbitals on the row,
- repeat on smaller M with the full self-consistent Σ_M^K ,
- compare the marker collapse against the corresponding spectral-function evolution at the interface.

000_rtriv3p000_M100_Nkx41_Gm0p500_y0to0 interface sum

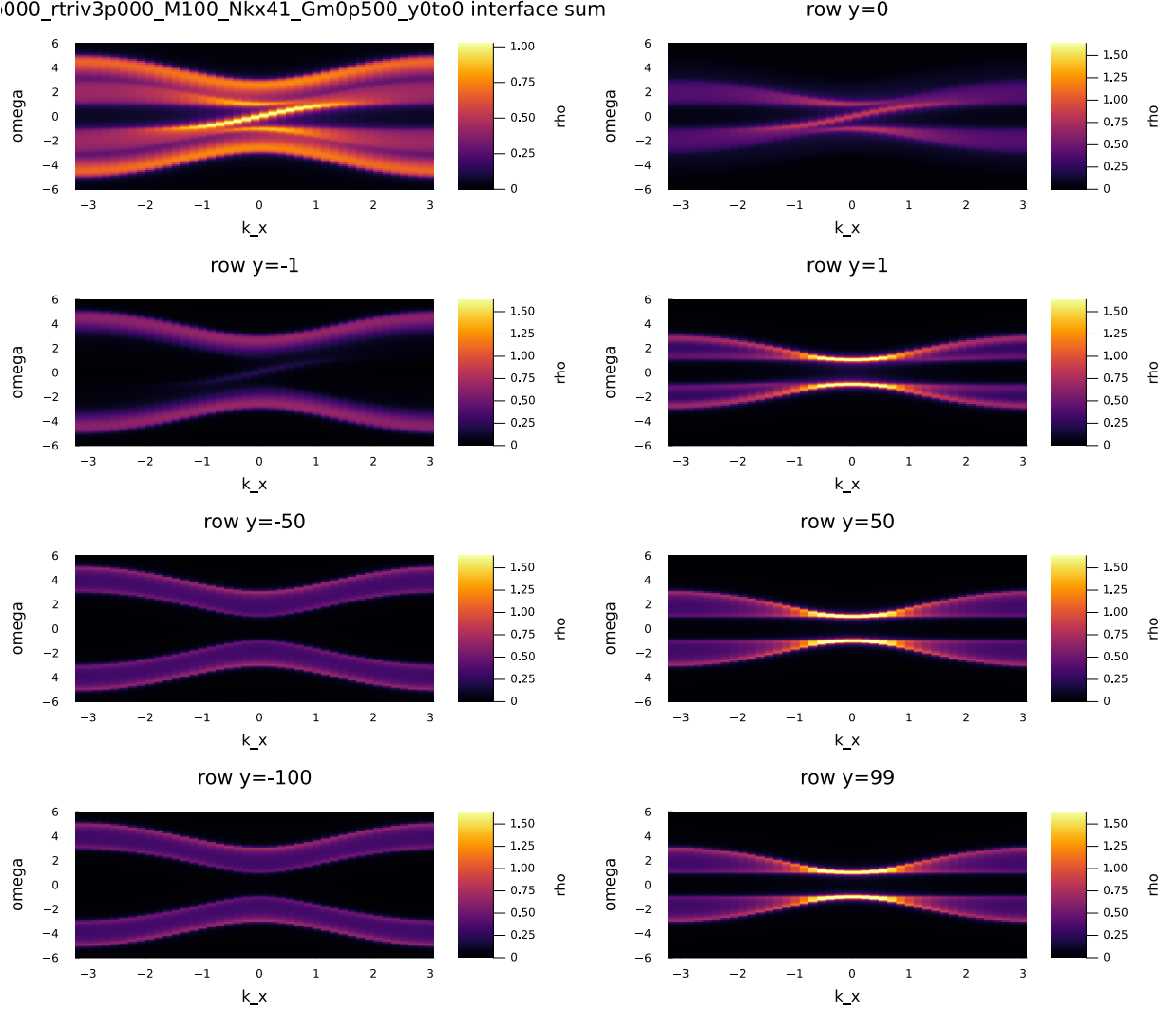


Figure 4: Monitored spectral output for the $M = 100$ slab at $\Gamma = 0.5$, with only the row $y = 0$ monitored and the half-filled monitored saddle imposed. As in the equilibrium case, the first panel is the interface-window sum, while the row-resolved spectral panels are arranged as $y < 0$ on the left and $y \geq 0$ on the right with a shared row color scale.

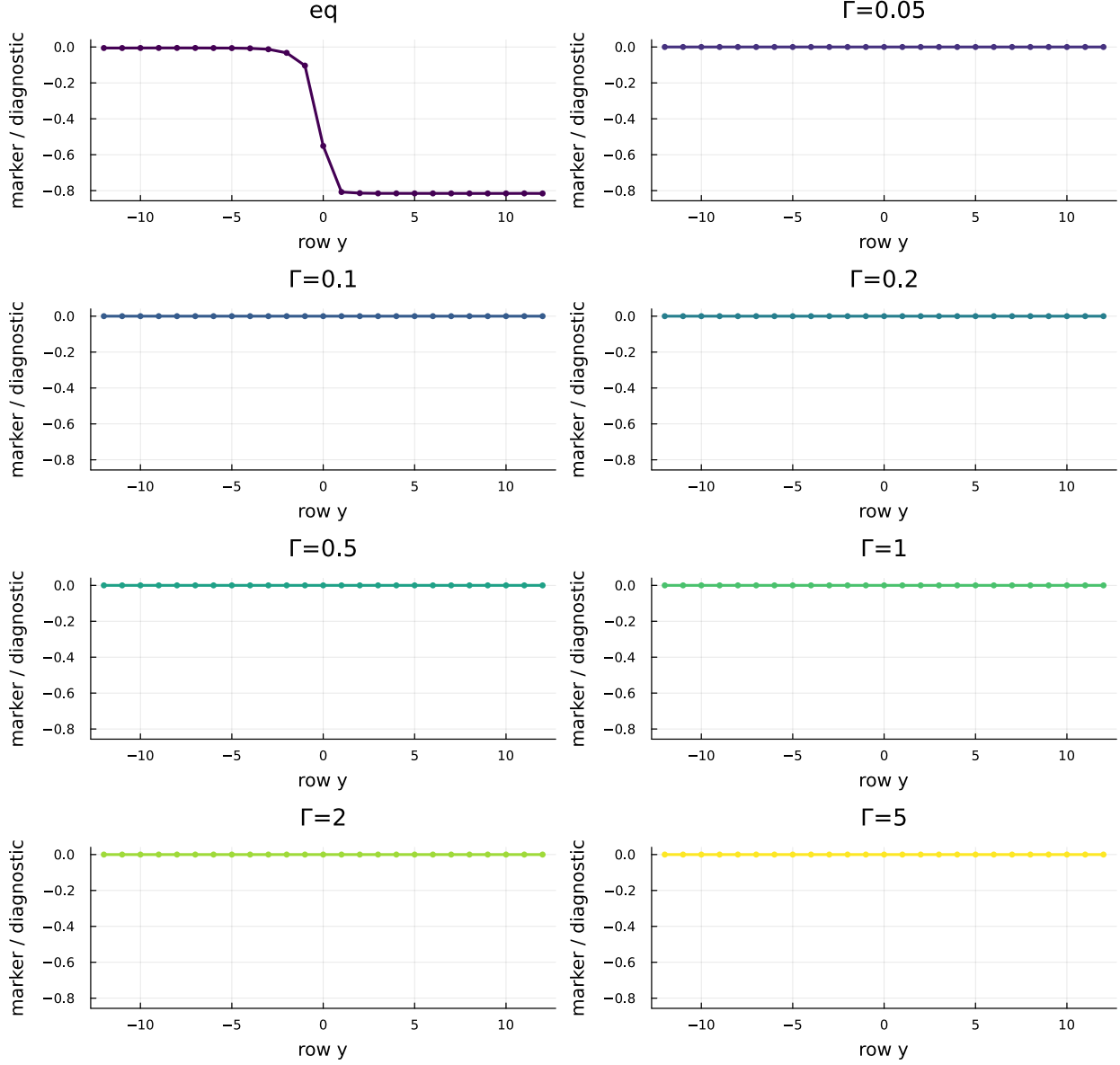


Figure 5: Panelized Γ -sweep for the $M = 100$ slab. The equilibrium panel shows the nontrivial interface profile. All monitored panels in the current setup are already collapsed near zero, even for $\Gamma = 0.05 \ll t_0$.